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Relationship between Poisson's ratio and soil suction for unsaturated soils

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ABSTRACT: In the present study, the variation of Poisson's ratio, ν with respect to degree of saturation, S for unsaturated soils is investigated. The maximum elastic, shear and constraint modulus data available in the literature measured using the bender element testing (*BET*) technique was used for the analysis. The Poisson's ratio, ν at different degrees of saturation, S were back-calculated using relationship between the elastic and shear modulus and the elastic and constraint modulus that were originally developed for homogeneous, isotropic, and linear elastic continuum. The results of the analysis show that the Poisson's ratio, ν is a function of degree of saturation, S and its characteristic behavior is closely related to the three phases of the Soil-Water Characteristic Curve (i.e. *SWCC*). In addition, the influence of Poisson's ratio, ν on the stress versus settlement behavior of unsaturated fine-grained soils below a shallow foundation is also studied from the Finite Element Analysis (*FEA*).

KEYWORDS: suction; unsaturated soils; Poisson's ratio; shear modulus; elastic modulus; shallow foundation

1 INTRODUCTION

One of the key parameters in the estimation of the stress versus settlement behavior of soils is Poisson's ratio, ν . This ratio is typically assumed to be constant in numerical modelling studies of both saturated and unsaturated soils. Such an assumption is reasonable for coarse-grained soils since the change in mean effective stress and associated volume change due to the wetting and drying cycles is negligible. However, in the case of fine-grained soils, the drying and wetting cycles can lead to the significant change in volume of soils, which results in different compressibility characteristics with respect to degree of saturation. This implies that the Poisson's ratio, ν of fine-grained soils is a function of degree of saturation, S (or soil suction, $(u_a - u_w)$).

In the present study, the variation of Poisson's ratio, ν with respect to the degree of saturation, S is investigated. Two sets of data from the literature (Mendoza et al. 2005 and Alramahi et al. 2010) were used to back-calculate the Poisson's ratio, ν for different degrees of saturation, S values using mathematical relationships between the elastic and shear modulus and the shear and constraint modulus originally developed for homogeneous, isotropic, and linear elastic continuum.

The study shows that the Poisson's ratio, ν is a function of degree of saturation, S and its characteristic behavior is closely related to the three phases of

the Soil-Water Characteristic Curve (i.e. boundary effect, transition effect, and residual zone) (Vanapalli et al. 1999). In addition, the influence of Poisson's ratio, ν on the stress versus settlement behavior of unsaturated fine-grained soils below a shallow foundation is also studied from the Finite Element Analysis (*FEA*). The analysis results showed that the applied stress versus surface settlement behavior of fine-grained soils is affected by not only Poisson's ratio, ν but also total cohesion, c .

2 VARIATION OF SMALL-STRAIN STIFFNESS OF UNSATURATED SOILS WITH RESPECT TO MATRIC SUCTION

2.1 Mendoza et al. (2005)

Mendoza et al. (2005) carried out bender element tests (*BET*) to study the influence of degree of saturation (i.e. suction) on the elastic and the shear modulus of commercial kalolinite clay ($w_L = 80\%$, $I_p = 35\%$, $G_s = 2.61$) for both drying and wetting paths. The elastic and shear modulus at small-strain (i.e. maximum elastic and shear modulus) from *BET* can be computed using Eq. (1) and Eq. (2), respectively.

$$E_{max} = \rho \cdot V_L^2 \quad (1)$$

$$G_{max} = \rho \cdot V_s^2 \quad (2)$$

where E_{max} , G_{max} = maximum elastic and shear modulus, ρ = specimen mass density; and V_L , V_S = L - and S -wave velocity.

They established relationship between the degree of saturation and suction using compacted specimens. The volume of the specimens was allowed to change during both drying and wetting process. Figure 1(a) and (b) shows the variation of E_{max} and G_{max} with respect to degree of saturation, respectively for four different specimens.

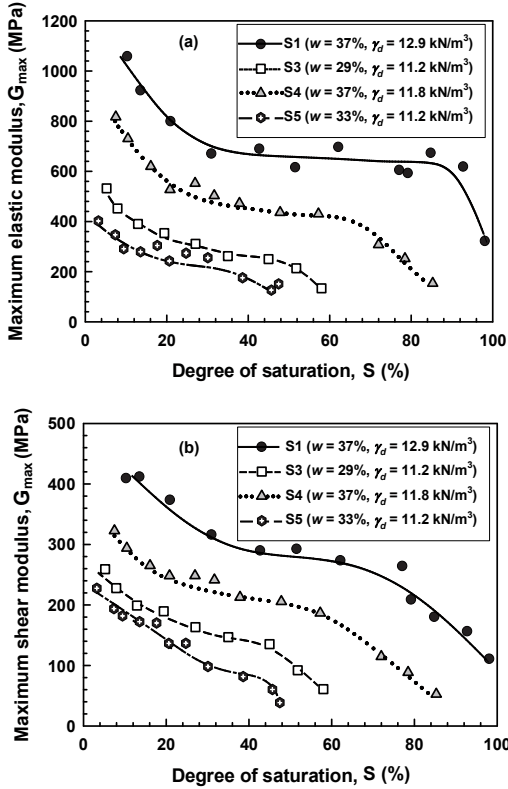


Figure 1. Variation of maximum (a) elastic (E_{max}) and (b) shear (G_{max}) modulus with respect to degree of saturation, S (data from Mendoza et al. 2005).

Mendoza et al. (2005) provided the explanation for the characteristic behavior of the E_{max} and G_{max} with respect to suction as below (Figure 2(b)).

- **Boundary effect zone:** sharp increase of E_{max} and G_{max} due to the increase in the density of the soil
- **Transition zone:** gradual increase of E_{max} and G_{max} due to some shrinkage
- **Residual zone:** sharp increase of E_{max} and G_{max} due to the particle displacement and rotation without major volume change

Based on the experimental results, they proposed an empirical equation to estimate the variation of E_{max} and G_{max} with respect to matric suction taking account of volume change as shown in Eq. (3) and Eq. (4), respectively.

$$E_{max(unsat)} = 30000 \frac{(2.3-e)^2}{1+e} [\ln(u_a - u_w)]^{1.35} \quad (3)$$

$$G_{max(unsat)} = 13300 \frac{(2.3-e)^2}{1+e} [\ln(u_a - u_w)]^{1.35} \quad (4)$$

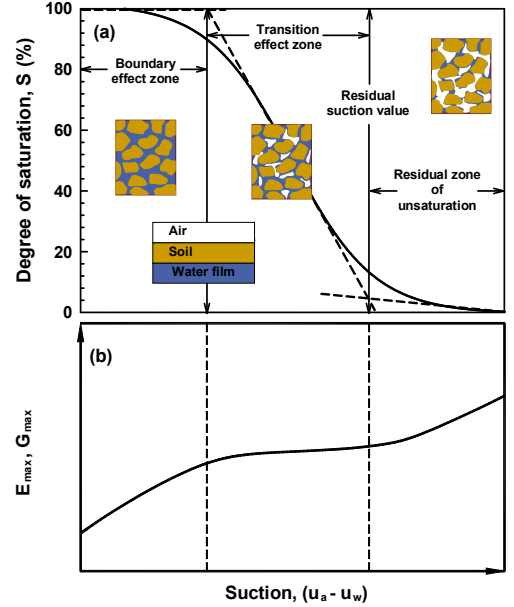


Figure 2. (a) Typical Soil-Water Characteristic Curve (SWCC) with three different stages of desaturation and (b) variation of elastic (E_{max}) and shear (G_{max}) modulus with respect to soil suction, $(u_a - u_w)$.

2.2 Alramahi et al. (2010)

Alramahi et al. (2010) carried out *BET* using a specially designed drying cell to study the variation of shear, constraint and bulk modulus with respect to degree of saturation. The samples were prepared by mixing spherical glass beads ($e = 0.51$, $\gamma_d = 17.24$ kN/m³, $C_u = 1.54$, $C_c = 0.87$) with three different pore fluids; i) deionized water, ii) deionized water with silt (12% silt by weight; $e = 0.52$, $\gamma_d = 17.17$ kN/m³), and iii) deionized water with kaolin clay (12% kaolin clay by weight; $e = 0.51$, $\gamma_d = 17.26$ kN/m³). The shear modulus was computed using Eq. (2) and the constraint and bulk modulus were computed using Eq. (5) and Eq. (6), respectively. In the present study, G_{max} and M_{max} values for the spherical glass mixed with clay and silt were used to investigate the variation of Poisson's ratio with respect to degree of saturation.

$$M_{max} = \rho \cdot V_p^2 \quad (5)$$

$$B_{max} = M_{max} - \frac{4}{3} G_{max} \quad (6)$$

where M_{max} , B_{max} = maximum constraint and bulk modulus; and V_p = P -wave velocity.

Figure 3(a) and (b) shows the measured G_{max} and M_{max} with respect to degree of saturation, respectively for the specimens of deionized water + clay (i.e. clay-slurry pore) and deionized water + silt (i.e. silt-slurry pore). Unlike the experimental results by Mendoza et al. (2005), a sudden drop of G_{max} and M_{max} is observed for zero degree of saturation. They suggested that this phenomenon can be attributed to the loss of capillary forces when soils completely desaturate.

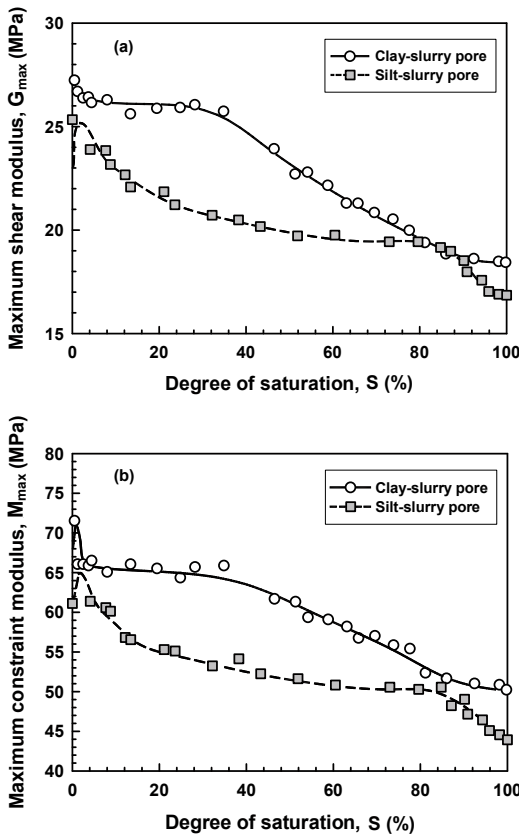


Figure 3. Variation of maximum (a) elastic (E_{max}) and (b) constraint (M_{max}) modulus with respect to degree of saturation, S (data from Alramahi et al. 2010).

Cho & Santamarina (2001) also provided explanations for possible reasons for the abrupt drop of G_{max} as a soil approaches to a fully dry condition (see curve ① in Figure 4). They conducted *BET* for four different specimens; i) glass beads (spherical parti-

cles), ii) a mixture of glass beads and kaolinite (clay), iii) a mixture of glass beads and granite power (silt) and Sandboil sand. Unlike the experimental results by Mendoza et al. (2005), the abrupt drop of G_{max} for the degree of saturation close to zero was not observed for the mixtures and Sandboil soil. However, such a behavior was observed for glass beads. They suggested that this behavior is attributed to the fact that the menisci between glass beads separated by a short distance fail earlier than those are in contact, which leads to the gradual decrease of the E_{max} and the G_{max} beyond optimum degree of saturation (i.e. degree of saturation corresponding to the highest E_{max} or G_{max} value). This explanation may not be applicable to the behavior of curve ② in Figure 4 (i.e. Clay-slurry pore in Figure 3) since the small-strain stiffness increases up to peak value at $S = 0$ as drying proceeds beyond point A (in Figure 4) due to the increase in bonding between small soil particles (i.e. fine-grained soil) (Cho & Santamarina 2001). The behavior such as curve ① in Figure 4 is generally observed for i) the uniformly graded coarse-grained soils (i.e. high coefficient of uniformity) which have relatively larger voids in the structures (Mendoza & Colmenares 2006) or ii) freshly remolded coarse-grained soils prepared at predetermined degree of saturation (Wu et al. 1984, Qian et al. 1993). However, if coarse-grained soils desaturate gradually by air drying the bonding between particles can increase continuously up to dry condition due to the effects that are similar to cementation (Rinaldi et al. 1998).

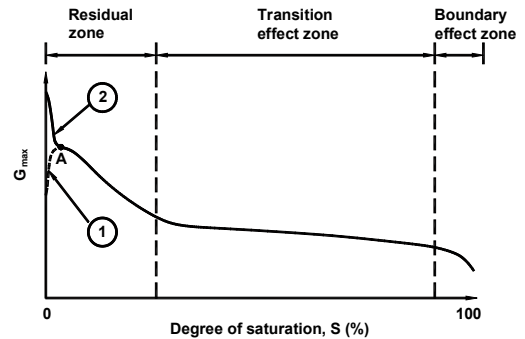


Figure 4. Variation of maximum shear modulus, G_{max} with respect to degree of saturation, S (modified after Cho & Santamarina 2001).

3 VARIATION OF POISSON'S RATIO WITH RESPECT TO SUCTION

The $E_{max} - G_{max}$ and $E_{max} - M_{max}$ relationship for homogeneous, isotropic, and linear elastic continuum are given in Eq. (7) and Eq. (8), respectively.

$$E_{max} = 2G_{max}(1 + \nu) \quad (7)$$

$$E_{max} = M_{max} \frac{(1+\nu)(1-2\nu)}{(1-\nu)} \quad (8)$$

Eq. (7) and Eq. (8) indicates that E_{max} can be computed if either G_{max} or M_{max} is known (i.e. measured) along with Poisson's ratio. In the present study, the concept in Eq. (7) and Eq. (8) is extended to E_{max} , G_{max} , and M_{max} values measured for unsaturated soils as shown in Eq. (9) and Eq. (10), respectively.

$$E_{max(unsat)} = 2G_{max(unsat)}(1+\nu) \quad (9)$$

$$E_{max(unsat)} = M_{max(unsat)} \frac{(1+\nu)(1-2\nu)}{(1-\nu)} \quad (10)$$

where, $E_{max(unsat)}$, $G_{max(unsat)}$, and $M_{max(unsat)} = E_{max}$, G_{max} , M_{max} for unsaturated condition.

To check the validity of Eq. (9) and Eq. (10), the $E_{max(unsat)}$, $G_{max(unsat)}$, and $M_{max(unsat)}$ values measured with BET technique for different degrees of saturation provided by (Mendoza et al. (2005) and Alramahi et al. (2010) were reanalyzed.

Figure 5(a) to (d) show the comparison between the measured E_{max} values and those estimated using Eq. (9) for different Poisson's ratio (i.e. $\nu = 0.1, 0.3$ and 0.5) for the data by Mendoza et al. (2005). There is a good agreement for $\nu = 0.1$ for entire range of degree of saturation. Lee & Santamarina (2005) suggested that $\nu \approx 0.1$ for unsaturated soils. Similar trends in results can also be observed when Poisson's ratio is back-calculated using Eq. (3) and Eq. (4) as below.

$$\nu = \frac{E_{max(unsat)}}{2G_{max(unsat)}} - 1 = 0.13 \quad (11)$$

This phenomenon is attributed to the fact that the influence of Poisson's ratio E_{max} (or G_{max}) for relatively high degrees of saturation (i.e. low suction values) is less compared to low degrees of saturation (i.e. high suction values).

To further examine the relationship between Poisson's ratio and degree of saturation, the Poisson's ratio was back-calculated and plotted against degree of saturation. Figure 6 shows the variation of Poisson's ratio with respect to degree of saturation for the data by Mendoza et al. (2005). The negative values and values greater than 0.5 of Poisson's ratio were neglected since they are not realistic. A trend line is drawn assuming $\nu = 0.1$ representing the relatively dry condition (low degree of saturation and high suction) based on the comparison between the measured and the estimated E_{max} values as discussed above (i.e. Figure 5(a) - (d), Eq. (11), which is also supported by Lee & Santamarina 2005). The Poisson's ratio values are almost constant with the values close to 0.5 in boundary effect zone, which indicates that the soil specimens were under undrained condition.

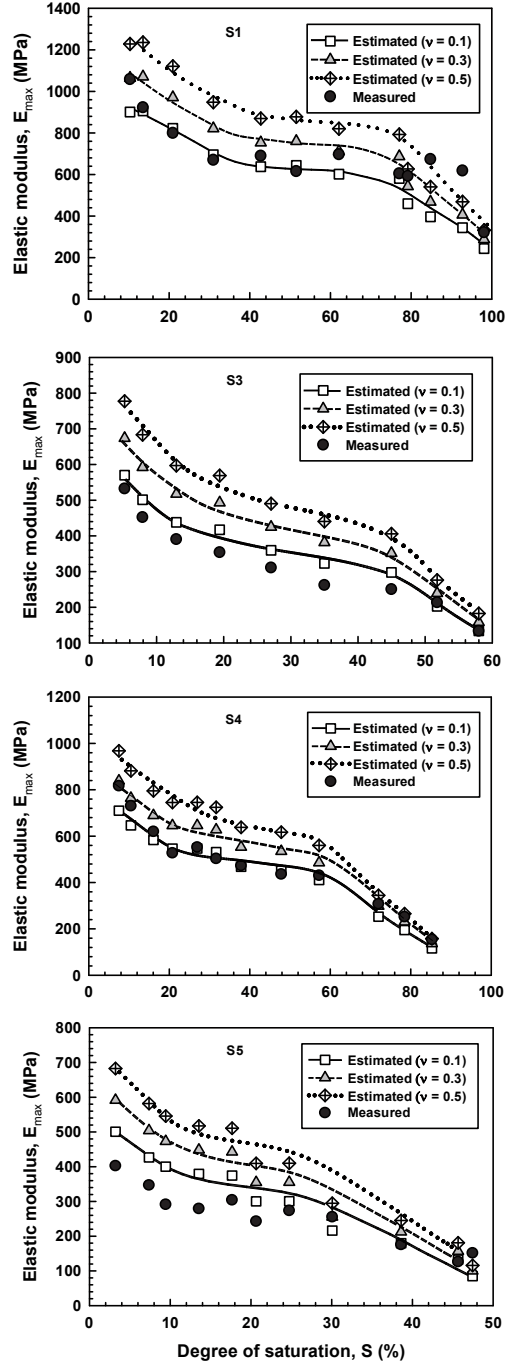


Figure 5. Comparison between the measured and the estimated maximum elastic modulus, E_{max} for different Poisson's ratio, ν (analyzing data from Mendoza et al. 2005).

The Poisson's ratio decreases with decreasing degree of saturation in the transition effect zone and eventually converges to a constant value in residual zone. The characteristic behavior of Poisson's ratio

with respect to degree of saturation implies that the variation of Poisson's ratio with respect to degree of saturation is closely related to the *SWCC*. In other words, the Poisson's ratio can be expressed as a function of degree of saturation (i.e. $f(S)$) and *SWCC* can be used as a tool to estimate or predict the variation of Poisson's ratio with respect to degree of saturation.

The discussion above implies that the Poisson's ratio is not constant but can vary with degree of saturation of soils. Therefore, Eq. (9) and Eq. (10) can be rewritten as below.

$$v^* = \left[\frac{E_{\max(\text{unsat})}}{2G_{\max(\text{unsat})}} \right] - 1 = f(S) \quad (12)$$

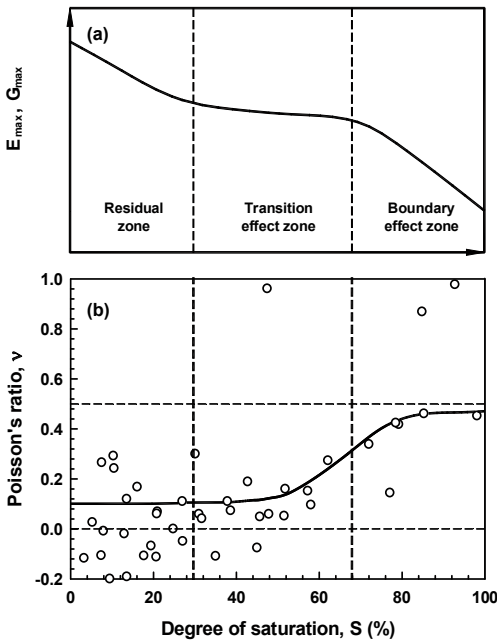


Figure 6. (a) Degree of saturation, S versus maximum elastic ($E_{\max}$) and shear ($G_{\max}$) modulus relationship and (b) back-calculated Poisson's ratio, v with respect to degree of saturation, S (analyzing data from Mendoza et al. 2005).

Figure 7(a) and (b) shows the variation of back-calculated Poisson's ratio with respect to degree of saturation for clay-slurry pore and silt-slurry pore, respectively provided by Alramahi et al. 2010. The behavior shown in Figure 7(a) (i.e. clay-slurry pore) is similar to that shown in Figure 6(b); however, this behavior is not observed for the silt-slurry pore specimen. The difference in the back-calculated Poisson's ratio values between the saturated (i.e. $S = 100\%$) and the dry (i.e. $S = 0\%$) conditions for clay specimen (Mendoza et al. 2005) and clay- and silt-slurry pore specimens (Alramahi et al. 2010) are found to be 0.4, 0.04, and 0.02 respectively. This fact

indicates that the variation of Poisson's ratio with respect to degree of saturation is more predominant for finer soil materials.

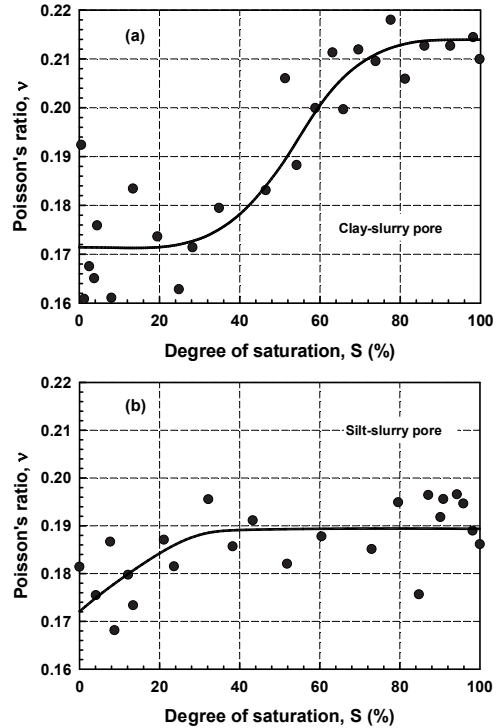


Figure 7. Back-calculated Poisson's ratio, v with respect to degree of saturation, S for (a) clay-slurry pore and (b) silt-slurry pore (analyzing data from Alramahi et al. 2010).

4 MODELLING THE STRESS VERSUS SETTLEMENT BEHAVIOR OF SHALLOW FOUNDATION

Finite element analysis (i.e. *FEA*) was carried out to model the stress versus settlement behavior for a shallow foundation located on unsaturated fine-grained soils. The Poisson's ratio with respect to degree of saturation data from Mendoza et al. (2005) is used in the analysis. The commercial finite element software, Geostudio (Ver. 2007, Geo-Slope Int.) was used for the *FEA*. The details of the how this analysis can be undertaken are described in Krahn (2007). The analysis was conducted using elastic-perfectly plastic model for $S = 60\%$ (i.e. $G_{\max} = 690$ MPa from *BET*) with different Poisson's ratio (i.e. $v = 0.25$ (back-calculated), 0.3, 0.35, 0.4, 0.5). The total cohesion was calculated using the semi-empirical model proposed by Vanapalli et al. (1996) and Fredlund et al. (1996) as given below.

$$c = c' + (u_a - u_w) (S^k) \tan \phi' \quad (13)$$

where, c = total cohesion, c' = effective cohesion; and κ = fitting parameter.

In the present study, $\kappa = 2.5$ ($I_p = 35\%$) was used to estimate the total cohesion based on the study by Garven & Vanapalli 2006. The total cohesion c (Eq. (13)), measured $G_{max(unsat)}$, and the estimated $E_{max(unsat)}$ values for different Poisson's ratio (Eq. (9)) are summarized in Table 1.

Table 1. Summary of the parameters used for the FEA.

$S(\%)$	c (kPa)	$G_{max(unsat)}$ (MPa)	ν	$E_{max(unsat)}$ (MPa)
60	245	273	0.25	670
			0.30	710
			0.35	738
			0.40	765
			0.50	820

Figure 8 shows the applied stress versus surface settlement behaviors obtained from the FEA for different $E_{max(unsat)}$ value (i.e. calculated with different Poisson's ratio using Eq. (9)). The study shows the difference in the settlement under the same applied stress is negligible regardless of the Poisson's ratio values. This is mainly attributed to the significantly high total cohesion (i.e. 245 kPa) value. In other words, the influence of Poisson's ratio on the applied stress versus surface settlement behavior under a shallow foundation can be neglected when degree of saturation is low and the total cohesion is relatively high. Figure 9 shows the variation of settlement with respect to Poisson's ratio under the same stress. The settlement is approximately the same for different Poisson's ratio values under a constant applied stress.

5 CONCLUSIONS

In the present study, the variation of Poisson's ratio, ν with respect to degree of saturation, S is investigated using the measured elastic, shear and constraint modulus from bender element tests available in the literature. The results show that the Poisson's ratio, ν is a function of degree of saturation, S . This characteristic behavior or relationship is closely related to the Soil-Water Characteristic Curve (SWCC).

FEA were performed to simulate the applied stress versus surface settlement behavior of a shallow foundation taking into account the variation of Poisson's ratio, ν with respect to the degree of saturation, S . The results of the study suggest that the influence of Poisson's ratio, ν on the applied stress versus surface settlement behavior is negligible when the degree of saturation, S is low. Such a behavior can be attributed to high total cohesion. However, more studies are necessary in this direction to study the influence of Poisson's ratio, ν on different types of stress versus displacement behaviors of unsaturated fine-

grained soils for relatively low values of degrees of saturation.

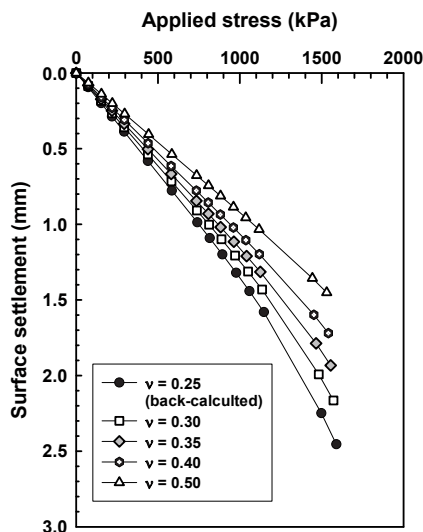


Figure 8. Applied stress versus surface settlement behavior for different Poisson's ratio, ν from the FEA.

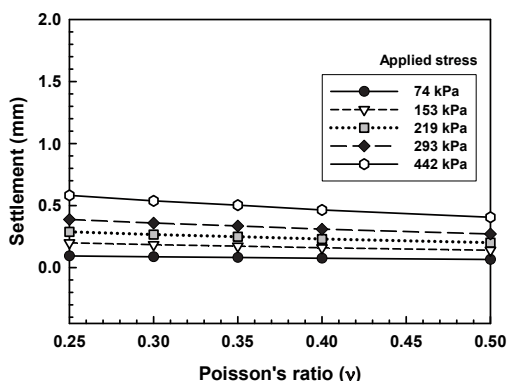


Figure 9. Variation of settlement with respect to Poisson's ratio under the same stress.

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